

common::ground: A Gamified Toolkit for Facilitating Meaningful Interactive Sessions

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Abstract

This paper introduces *common::ground*, a flexible toolkit that assists facilitators in conducting more engaging, meaningful, inclusive, and sustainable icebreaking sessions for culturally diverse groups. Icebreakers play a crucial role in fostering initial social cohesion, but many existing solutions fail to engage participants from diverse backgrounds, leading to superficial conversations or discomfort due to language barriers and mismatched expectations. Using a mixed-methods research process, we conducted secondary research, collected survey (n=19) and interview (n=8) data, refined prototypes through iterative design, and performed usability studies (n=9) in an academic setting. Our findings demonstrate that structured resources, progressive conversation prompts, and facilitator-centered support significantly improve participant comfort and engagement. Usability study revealed high design goals alignment, moderate to strong usability, and overall positive participant feedback. We aim to contribute to the established works in user experience and interactive sessions, offering a validated design intervention and a model for further explorations in cross-cultural participant engagement.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI); Interaction design.**

Keywords

Human-centered Design, Cross-Cultural Interaction; Icebreakers; Gamification; Cultural Exchange; Toolkit; Facilitator

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1 Introduction

Icebreakers are interactive sessions intended to ease participant transition into new educational or professional contexts [6]. Although widely used, many standard approaches (for instance, brief "fun facts" or simplistic name-chain activities) often fail to address crucial cross-cultural dynamics. Research indicates that mismatched norms, language barriers, and unfamiliar social cues can increase disengagement among diverse groups, resulting in shallow conversation and limited trust [3, 15]. These limitations are particularly pronounced in large academic cohorts or workshop settings where attendees may differ significantly in language proficiency, cultural conventions, and comfort levels with self-disclosure [13]. Previous attempts to design more inclusive icebreakers, such as digital games [10] or large shared displays [11], have made progress but frequently remain technologically heavy or culturally non-specific, possibly alienating or confusing participants lacking advanced fluency or technical familiarity.

In cross-cultural environments, facilitation is a key factor: group facilitators who receive guidance on structuring comfortable, meaningful, and supportive sessions can substantially increase engagement and inclusion. Studies highlight that language barriers do not solely stem from vocabulary deficits— conversation pacing, turn-taking norms, humor styles, and cultural references can all obstruct participants from forging deeper bonds [1].

The motivation for developing a facilitator-centered toolkit that addresses these concerns emerged from our surveys and interviews, where facilitators and participants expressed frustration with superficial questions, insufficient or confusing guidance, and minimal recognition of cultural identities. These initial findings support the broad claim in user experience and usability studies that design interventions informed by iterative, human-centered design are more likely to yield effective solutions [2, 5]. In line with this paradigm, we employed a mixed-methods research aimed at enabling facilitators to facilitate sessions that engage participants at a comfortable level in cross-cultural settings.

By designing a toolkit with tangible artifacts like Bingo Boards, Interaction Cards, Name Tags, Stickers, dICE, and Facilitator Support guidelines, we propose *common::ground*. This paper details its conceptual underpinnings, design decisions, and evaluation outcomes from a usability study. Our central argument rests on the premise that designing for inclusive engagement, especially in culturally diverse contexts, demands a synergy of structured prompts, minimalist technology usage, and a consistent design language that encourages personal sharing without compelling forced sharing.

Our research question going into this study is: How can a gamified toolkit support facilitators in facilitating icebreaker sessions that feel engaging, flexible, meaningful, inclusive, and sustainable in cross-cultural settings?

In the following sections, we first position our work in the broader background of facilitator-centered design, gamification, cross-cultural communication, pertaining to small to moderate-sized interactive sessions. Through this lens, we then document our human-centered design approach, highlighting the design directions, goals, and decisions made with each iteration. We outline the usability study that guided the final high-fidelity prototype of *common::ground*, followed by a discussion of the results and limitations. We define future work directions consistent with our design ethos, and ultimately reaffirm the need for a well-integrated, minimal-technology approach that enables conversations in cross-cultural settings.

Following the value of Humanistic HCI [2], we present a disclosure to our readers of the conditions of our own knowing. The authors of this paper are from three different countries and are studying Human Centered Design and Engineering at the University of Washington. We all have firsthand experience with icebreakers in cross-cultural settings. This background information might help readers understand where we are coming from and why we would collaborate on this together.

2 Background

Previous works in Human-Computer Interaction and Computer-Supported Cooperative Work have explored playful designs to promote initial social bonding [3, 6, 15]. Studies show that participants often benefit from structured prompts, especially in environments where cultural references or linguistic fluency differ significantly [9, 10]. Although some research has integrated digital or large-display elements to lower apprehension [3, 6], these approaches are not always suitable for smaller indoor contexts or speed-critical moments (e.g., short orientation sessions).

The following subsections serve as a basis to generate our design directions, covering facilitator-centered design, gamification, language barriers, and cross-cultural communication.

2.1 Facilitator-Centered Design

While many references in HCI discuss using games or playful technology for group formation [9, 15], fewer highlight the preparatory or adaptive role of the facilitator. Facilitators often handle group goals, time constraints, participant comfort, and cultural sensitivity. Observational evidence suggests that well-structured instructions and flexible "activity modules" can help novices manage these complex demands [3, 6].

2.2 Gamification in Social Interaction

Gamification in social settings can be beneficial by providing structure and lighthearted competitiveness, so long as these mechanics are accessible, fair, and context-appropriate [6, 10]. Bingo, for instance, has historically been used in language-learning contexts to build familiarity with new vocabulary, while its flexible rules can be adapted for open-ended conversation [15]. Critics of over-gamification caution that scoring systems and oversimplified "achievement" icons can trivialize personal experiences or overshadow deeper group bonding [7]. Designing a measured approach that triggers conversation while prioritizing relationship-building becomes critical—particularly in intercultural contexts that demand sensitivity and clarity [7, 10].

2.3 Language Barriers

Language barriers exceed superficial vocabulary mismatches—they encompass concerns about accent recognition, pacing, chosen topics, and high-stakes misunderstandings amplified by group contexts [15]. For instance, participants may refrain from speaking to avoid revealing an accent, while native speakers might fail to check for comprehension. Cross-cultural design solutions must minimize forced disclosure, clarifying instructions in multiple modes (visual, textual, or symbolic) and offering prompts that adapt to different comfort levels. Moreover, social and cognitive scientists have shown that inclusive group tasks can enhance empathy and reduce stereotypes [9], underscoring how the solution design might cultivate deeper social learning once participants feel psychologically safe.

Language barriers affect comprehension, pacing of talk, and willingness to participate, while cultural differences can influence humor norms, conversation topics, and self-disclosure preferences [10, 15]. In our survey (n=19), 66.7% of respondents mentioned language barriers, and 50.0% cited difficulties adapting to alternative communication styles. Participants felt excluded or forced to "fake" shared experiences that did not match their cultural contexts.

2.4 Cross-Cultural Communication and Comfort

Studies in small-group communication highlight that comfort in early sessions correlates with future collaborative success, with trust forming faster when structured tasks guide turn-taking and topic selection. Cultural mismatch complicates the formation of that trust, as norms of non-verbal expression or personal boundaries vary by region. Minimizing anxiety through gentle, culturally neutral prompts can alleviate tension among participants who share neither first-language proficiency nor certain background references, such as local sports or brand knowledge [13]. Intentionally curated topics (e.g., generalizable questions about personal hobbies or early-life experiences) help unify participants and avoid cornering them into culturally exclusive trivia in the initial stages of conversations [3, 11].

2.5 Design Directions for *common::ground*

Consolidating our secondary research findings, we derived three key design directions. First, the solution must center around the facilitator, offering them tools to adapt to group size, available time, or emergent participant needs. Second, the system should facilitate

deeper conversation gradually, without imposing forced personal revelations or other potentials for discomfort or embarrassment. Third, materials must be straightforward and supportive of a wide range of language proficiencies and cultural backgrounds, aiding to overall participation comfort and engagement.

3 Toolkit Design

3.1 Methodology

The design of the *common::ground* toolkit followed a human-centered design approach, with data-driven changes made over iterations. Our team triangulated findings from secondary research, need identification surveys (n=19) and interviews (n=8) for the iterative design and prototyping of the toolkit.

Our initial design directions navigated us to five key design goals: Engaging, Flexible, Meaningful, Inclusive and Sustainable. Following an ideation and critique session, we designed a low-fidelity prototype featuring Passport booklets comprising Bingo Boards and guidelines for facilitators and participants, Interaction Cards, basic Name Tags, and a limited variety of Stickers. Pilot testing on this toolkit prototype revealed user needs for more robust prompts and clearer instructions, leading us towards the mid-fidelity design of Passports, tiered (prompt comfort levels and categories) Interaction Cards, and more varied and inclusive Sticker sets.

Further design refinements were guided by usability study of our mid-fidelity prototype. We tested the clarity of instructions— both for facilitators and participants. As we discovered that many participants felt overwhelmed or uncertain during the initial stages, we added structured guidance for facilitators and simplified the instructions. Our final, high-fidelity prototype consists of the following components. The specifications for the functional components of the *common::ground* toolkit, including their constraints, technologies, variables, and actions, are shown in the Token and Constraints (TAC) [12] Palette table 1.

3.2 Final Toolkit Components

3.2.1 Bingo Boards: (Fig. 1. A)

Bingo Boards incorporate multiple formats: a standard 5×5 version for full sessions, plus a 1×3 tutorial row. Reward systems, such as special stickers, further reinforce participation and motivate engagement. Overall, this component is not a required mechanism to boost participant engagement, but a flexible addition, giving facilitators the agency to incorporate dynamics of gamification and rewards in their sessions.

3.2.2 Interaction Cards: (Fig. 1. B)

Interaction Cards include five **standard** categories (Culture, Academics, Career, About Me, Surprise!), and a DIY (Do-It-Yourself) category where participants fill their own cards, with progressive levels (1–5) ranging from baseline engagement to personal sharing, to cater to varying degrees of comfort.

3.2.3 Name Tags: (Fig. 1. C)

We designed language-based Name Tags after our pilot testing revealed that participants felt represented seeing their representative languages. Design composition was optimized based on user feedback on readability, visibility, and effective placement of stickers.

3.2.4 Stickers: (Fig. 1. D)

We designed a more inclusive and representative sticker set based on our survey showing a need for diverse cultural expression (66.7% of survey respondents faced cultural barriers).

3.2.5 dICE: (Fig. 1. E)

The six sides represent card categories for randomized selection; based on the research team's ideas to expand the game mechanics. dICE also functions as a flexible addition to the toolkit, providing facilitators with the agency to incorporate another level of gamification in their sessions.

3.2.6 Facilitator Support: (Fig. 1. F)

A single consolidated guide with sample activity flow, quick setup checklist, key components and an emergency cheatsheet. Emergency guides were added based on frequently occurring conflict scenarios identified in usability study. Facilitator support is flexible: the guide expands as the facilitator gets familiar with the varied contexts and variables going into interactive sessions.

3.2.7 Toolkit Packaging: (Fig. 3. A)

A facilitator-centered packaging design was chosen based on survey findings showing preference for organized, professional presentation while maintaining portability.

3.3 Usage of *common::ground* Toolkit

Non-linear activity flows allow facilitators to switch between tools dynamically, maintaining session flexibility and engagement (Fig. 2. A, B, C). Facilitators are introduced to the *common::ground* toolkit with **Facilitator Support**. A sample activity flow explains how the constituting components work together dynamically to aid facilitation for an interactive session. A brief usage flow pertaining to a **facilitator as a reader** is as follows:

3.3.1 Welcome and Introduction: Briefly explain the toolkit and its components. Distribute materials to participants and allow time for onboarding and customization.

3.3.2 Icebreaking: Prompt participants to use Interaction Cards, starting with easier comfort levels, to ease them into the session. Encourage paired or grouped introductions.

3.3.3 Optional Gamified Additions: Participants work in pairs or groups to complete their Bingo Boards: for shorter sessions, focus on quick, casual exchanges (e.g., hobbies, fun facts). For longer sessions, prompt participants to progress through deeper levels (e.g., values, life goals).

3.3.4 Regular Check-ins: Circulate the session venue to monitor group dynamics. Address confusions and provide clarifications on components and their overall flow. Encourage participants to switch pairs or groups and interact with others, if time and session goals align.

3.3.5 Session Wrap-up and Reflection: Encourage participants to share takeaways or reflections. Prompt them to choose from take-home stickers or other memorabilia if a reward-based mechanism is included in the session.

Token	Constraints	Technologies	Variables	Actions
Bingo Boards	- Grid size limits cell size - Space for writing, marking - Board size (5x5) based on the number of standard categories (5)	Paper-based printing	- Board size - Bingo types (row, column)	Fill, Mark, Complete, Reward
Interaction Cards	- Physical card size limits text - Must fit in box - Level system (1-5) - Category system (5 (standard) + 1 (DIY) types)	Paper-based printing	- Difficulty levels - Categories	Draw, Perform, Replace
Name Tags	- Fixed sticker space - Visibility on clothing - Marker compatibility	Paper-based printing on white sticker paper with reusable adhesive	- Language options - Personal representation	Write, Customize, Wear, Store for future use
Stickers	- Size limits detail - Must fit on a name tag - Adhesive properties	Paper-based printing on sticker paper	- Categories - Quantity within categories	Select, Apply, Remove, Customize
dICE	- Six faces: number of categories shall match the number of card categories - Must be readable	Cardboard/Plastic	- Categories - Mechanism to accommodate mismatch in category count	Roll, Select, Match

Table 1: Overview of Tokens and Their Characteristics (TAC Palette) [12]



Figure 1: Components of the toolkit include: A) Bingo Board. B) Interaction Cards. C) Name Tags. D) Stickers. E) dICE. F) Facilitator Support.

4 Usability Study

4.1 Testing and Metrics

Our primary objective was to investigate the usability of our **mid-fidelity prototype** of the *common::ground* toolkit for facilitators and participants, to guide our final, high-fidelity prototype, as described in the **Toolkit Design** section above.

We employed convenience sampling [8], a non-probability sampling method where participants are selected based on ease of access and proximity, to recruit $n=9$ study participants.

The participants' ages ranged from early to late twenties, with five self-identifying as international and four as domestic students. All participants had prior experience with icebreaking and similar sessions. They had elementary facilitating experience. They demonstrated moderate social tendencies (introvert(1)-extrovert(5) scale: 3/5) and high English proficiency (4.625/5). The latter aligned with our expectations going into the study as English is our university's official medium of instruction, giving affiliated students one common language to interact.

We used roleplay testing [14], a method where participants assume specific roles to evaluate system performance in realistic scenarios. One participant was designated as facilitator to simulate

real-world usage by new facilitators, and the other eight were assigned to be student participants, our secondary users. Our research team maintained a passive observer role throughout the session, taking detailed observational notes.

The test was conducted in a small meeting room in our university, designed to simulate a first-day classroom icebreaking scenario. The session started after taking informed verbal consents from participants and lasted around 30 minutes. The session followed a structured format beginning with an initial briefing and material distribution. The facilitator led the introduction to toolkit components and guided participants through paired Interaction Cards activity. Participants engaged with the Bingo Boards in Passports, while researchers observed their interactions. A naturalistic element was introduced when one participant arrived late, allowing observation of the facilitator's impromptu onboarding process. The session remained primarily unmoderated, with the facilitator managing group dynamics and activity progression.

We designed a comprehensive mixed-methods evaluation approach combining quantitative metrics and qualitative insights to assess both usability and design goal achievement. Our analysis methodology included: 1) Design Goal Evaluation using 5-point

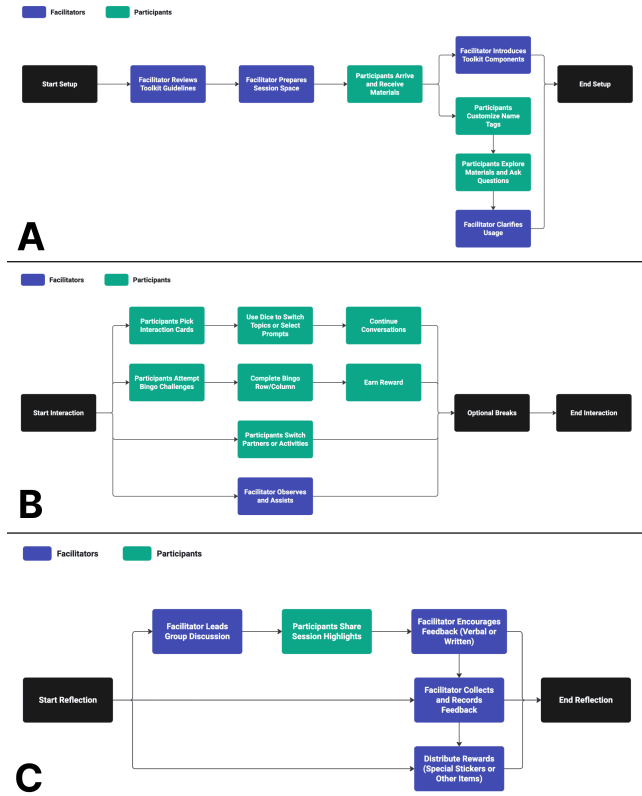


Figure 2: Sample Session User Flows: A) Setup- Prepare materials and establish the session environment. B) Interaction- Facilitate dynamic participant interactions using toolkit components. C) Reflection- Conclude the session with feedback, sharing highlights, and rewarding participants.



Figure 3: A) Toolkit in its packaging. B) Ongoing usability study. C) Post-session toolkit usage status.

Likert-scale items to assess five key metrics- Engagement (fun and rewarding experiences), Flexibility (adaptability across scenarios), Meaningfulness (conversation effectiveness), Inclusivity (social and emotional safety), and Sustainability (reusability and eco-friendliness); 2) System Usability Scale (SUS) questionnaire [4]; 3) Post-test semi-structured interviews to gather qualitative feedback; 4) Participant behavioral observations.

4.2 Data Analyses

For all the quantitative metrics, we evaluated the descriptive statistics and highlighted the mean scores across the Design Goal

Evaluation and System Usability Scale. For all the qualitative metrics, we categorized the collected insights by user type (Facilitator and Participants).

4.3 Quantitative Insights

Design Goal Evaluation (5-point Likert scale) results, as shown in tables 2 and 3, highlighted the following:

- Passport: Highest overall rating (4.25/5) among participants with standout performance (4.50/5) in the meaningfulness category.
- Interaction Cards: Highest overall rating (4.8/5) for the facilitator. Especially, four categories (Engagement, Flexibility, Meaningfulness, and Inclusivity) reached the maximum score.
- Name Tags and Stickers ranked third and fourth respectively as components for both facilitator and participants.
- Overall Toolkit as one system has strong performance (around 4.0/5) for both facilitator and participants.

The SUS Score for the **facilitator** was **72.5**, which is in the "Good" usability range, above the industry average of 68. The results showed that the toolkit is easy to use, of low complexity, and the facilitator was confident to use it. The average SUS Score for **participants** was **67.2** (SD=17.9) as in the range of "OK". The high standard deviation indicates substantial variation, with some participants rating the overall toolkit as extremely easy and fun, while others felt uncertain. This finding is further discussed in the following sections.

4.4 Qualitative Insights

4.4.1 Facilitator: The facilitator appreciated components in the toolkit. For Interaction Cards, the facilitator valued the diverse and engaging prompts as they fostered meaningful conversations among participants. Bingo Boards in Passports were considered "cute," and the game mechanics helped the facilitator moderate the whole session more easily. Turning to Name Tags, the facilitator valued their adaptability to serve for later conversation. Opportunities for design iteration also emerged. The facilitator struggled with understanding and conveying Bingo game mechanics to participants. The facilitator suggested that a more direct, step-by-step script/tutorial or a practice round would ease cognitive overload. Also, the facilitator mentioned the tension between deep conversation and Bingo game mechanics. The facilitator said, "We kept talking about things, and so sometimes we just forget 'Oh, wait. We are supposed to wrap it up.'" Despite the tension in Bingo game design, this also implies the lack of clear timing guidance for the facilitator.

4.4.2 Participants: Moderator observation and post-session interviews highlighted that participants generally appreciated the toolkit experience in helping them to find common grounds with others. P3 suggested his satisfaction and engagement with the overall experience by saying, "It's surprising that we have a lot of things that are actually in common, such as the things we think about the future and career, and favorite food." Similarly, P2 said, "We have very common thoughts about the things we need to teach ourselves if we had the

	Engaging	Flexible	Meaningful	Inclusive	Sustainable	Average
Passport (Bingo Board + Instructions)	4	4	4	4	4	4.0
Interaction Cards	5	5	5	5	4	4.8
Name Tags	3	4	4	4	4	3.8
Stickers	3	2	4	5	4	3.6
Overall Toolkit	4	3	5	4	4	4.0

Table 2: Design Goal Evaluation by the Facilitator.

	Engaging	Flexible	Meaningful	Inclusive	Sustainable	Average
Passport (Bingo Board + Instructions)	4.38	4.25	4.50	3.88	4.38	4.25
Interaction Cards	4.25	4.25	4.13	3.75	4.25	4.09
Name Tags	3.75	4.00	3.75	3.38	3.75	3.72
Stickers	4.00	3.88	3.50	3.00	4.00	3.59
Overall Toolkit	4.25	4.38	3.86	3.50	4.25	4.00

Table 3: Design Goal Evaluation by Student Participants. *The data presented above has been rounded up to two decimal places.*

chance, which is passion." Especially, category-based prompts offered tangible starting points for conversation. P2 appreciated those prompts, commenting, *"(They) are very good questions that really reflect our thoughts."* That is to say, prompts prepare participants with topics, which enable the following conversations to happen more easily and deeply.

Some tension was observed with the game mechanics and in-depth conversations as the participants used Bingo Boards. Roughly half the participants admitted they lingered on certain prompts without marking Bingo tiles. As they valued deeper discussion, the objective of the game's formal objective was overshadowed. P5 said, *"So we ended up covering very few questions compared to a lot of people who were just like 'bingo bingo bingo'."* This shows the competitive characteristics of the Bingo game mechanics, designed to inspire conversations in larger quantity, might restrict conversation in more depth.

Overall, the interplay of structured tasks and flexible personal expression was described by both facilitator and participants as *"good"* to initiate conversation for the icebreaking session. However, tension with design in different toolkit components suggest directions for future iteration.

5 Discussion and Future Works

There are many design opportunities to make *common::ground* attune to diverse user contexts. In very short sessions, it may be challenging for late-arriving participants to get familiar with instructions. Our current toolkit design requires participants to speak at least one common language. Future works can explore adding more contextually relevant and inclusive design elements into the toolkit, such as using more symbolic representation of prompts or multilingual translations.

While the usability study highlights promising outcomes, it was conducted in a single academic site and session with nine participants, limited in terms of settings, range of age and experience levels, and number of participants. Firstly, longitudinal studies could testify whether repeated usage across multiple sessions aligns with our stated design goals or whether participants eventually find the

repeated game mechanics stale. Also, the cultural mix of participants in our usability study was moderately diverse, but further testing across more varied age or experience level scales could reveal how well the toolkit usability translates to different contexts. Additionally, the current toolkit's potential to cater to large groups (e.g., 100+ attendees in a conference keynote) has not been tested yet. The results so far indicate success in medium-sized, classroom-like contexts (8–20 participants), though advanced facilitation methods might adapt the system to larger gatherings.

Close alignment with minimal-tech, face-to-face engagement remains a core design value, so expansions targeting adding digital interventions might consider how to incorporate those features without undermining personal interactions. We envision low-tech additions aligning with our toolkit's flexibility, such as printable expansions or modular add-on card decks for specific event types. We also see potential in cross-cultural conversation guidelines included in the Facilitator Support, clarifying how the facilitator might handle culturally sensitive topics or conflict situations.

Ultimately, future research could investigate the extent to which participants adopt new social connections beyond the session itself. This might involve short follow-ups to see whether shared interests lead to continuing interactions. Another possibility is embedding more representative frameworks of cultural identity. The aim is to refine a consistent design approach that upholds deeper human connection through tangible, culturally mindful, and gently guided conversations.

6 Conclusion

We have introduced *common::ground*, a facilitator-centered, gamified toolkit designed to support cross-cultural icebreakers to be more engaging, flexible, meaningful, inclusive, and sustainable. Grounded in secondary research, survey, interview, and usability study, the final toolkit prototype integrates tangible artifacts (Bingo Boards, Interaction Cards, Name Tags, Stickers, dICE, Facilitator Support). Usability study revealed our toolkit's appropriate fulfillment of our design goals, demonstrating moderate to high usability and overall positive participant feedback. Future explorations could

build upon our design ethos, methods, and opportunities for enabling better cross-cultural interactive sessions.

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